Free-Floating Rationales, Normative Hysteresis, and Constitutional Lock-in:

Dennett's Contribution to Extended Phenotype Theory and Multilevel Evolutionary Game Theory in Legal Systems

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March 2026

Abstract

This paper develops three interrelated contributions to the research program combining Extended Phenotype Theory (EPT) and Multilevel Evolutionary Game Theory (EGT) as applied to legal systems.

First, it integrates Daniel Dennett's philosophical apparatus (free-floating rationales, the cranes/skyhooks distinction, and sphexism) as the cognitive-mechanistic foundation connecting Dawkins's gene-centered memetics to Gould's hierarchical structuralism.

Second, it introduces the concept of normative hysteresis as a mechanism analytically distinct from path dependence, institutional lock-in, and architectural spandrels: while spandrels describe structural byproducts and the Constitutional Lock-in Index (CLI) measures current rigidity, normative hysteresis captures the cumulative, irreversible deformation that each normative intervention leaves in the institutional substrate, a deformation whose return path is never the inverse of the departure path.

Third, it develops the parallelism/convergence distinction drawn from evo-devo biology and applies it to legal transplant failure: transplants fail not because pressures are absent but because the recipient system lacks the deep homologous generators that the transplant presupposes. These three contributions generate five falsifiable predictions tested against the Argentine labor reform dataset (23 attempts, 1990–2025) and cross-national comparisons (Argentina CLI 0.89; Chile 0.24; Brazil 0.40; Spain 0.51).

The paper builds directly on Lerer (2026a, DOI: 10.5281/zenodo.18829975) and fills the causal gap that prior work identified but could not close: if legal actors deliberate, how can memetic selection operate blindly? Dennett's answer is that deliberation is real but operates at the organism level; selection operates at the replicator level through free-floating rationales that no one formulated and sphexistic routines that no one designed.

Keywords: Extended Phenotype Theory, Evolutionary Game Theory, Constitutional Lock-in Index, normative hysteresis, free-floating rationales, sphexism, evo-devo, legal transplants, Argentine labor law, memetics

Related DOIs: 10.5281/zenodo.18829975 (Multilevel Selection, CLI 3.0)

I. Introduction: Closing a Causal Gap

The research program associating Extended Phenotype Theory with the evolutionary analysis of legal systems has, since its initial formulation (Lerer, 2025a), generated a persistent methodological objection that its empirical results cannot deflect. The CLI achieves 78–82% predictive accuracy across 60 reform attempts in six Latin American countries. The dataset of 23 Argentine labor reforms between 1990 and 2025 shows a sustained failure rate consistent with CLI 0.89. The quantitative architecture is, by the standards of the field, unusual in its rigor. And yet the objection recurs: if jurists deliberate consciously, if legislatures reason about policy, if reform coalitions strategically calculate their moves, in what sense is there selection without a selector? In what sense can one speak of memetic fitness when every carrier of legal doctrine is a reflective agent capable of choosing otherwise?

Prior work (Lerer, 2026a) addressed this objection by integrating Gould's multilevel hierarchy with gene-centered memetics. The conclusion was that higher-level emergent effects are real but operate as downward constraints rather than selection units; that MLS2 (multilevel selection type 2, where groups are fitness-tracked as units) lacks causal microfoundations when applied to legal systems; and that the CLI mathematically formalizes Gould's developmental constraints through the Birkhoff polytope. What that paper did not resolve was the cognitive mechanism: how does constraint-without-consciousness operate in a domain populated by conscious agents?

Daniel Dennett closes that gap. His philosophical contributions across three decades of work on intentionality, cultural evolution, and consciousness provide the mechanism that Dawkins's memetics requires and that Gould's structuralism assumes without explaining. The present paper develops this integration in three stages. The first (Section II) deploys three Dennett concepts, each of which maps a specific explanatory lacuna in the EPT-EGT framework. The second (Section III) integrates these concepts with the Heteronomous Bayesian Updating model and the EGT typology of cooperative intentionality. The third (Sections IV and V) introduces two new theoretical contributions not present in prior work: normative hysteresis as a mechanism distinct from lock-in and spandrels, and the parallelism/convergence distinction from evo-devo as a diagnostic for legal transplant failure. Sections VI–VIII apply the resulting framework to the Argentine case and generate five falsifiable predictions.

II. Three Dennett Concepts and Their Legal Translation

2.1 Free-Floating Rationales: Reasons Without Authors

In Darwin's Dangerous Idea (1995), Dennett introduced the concept of free-floating rationales to describe an evolutionary design property: organisms embody reasons for their structure that no agent ever formulated. The giraffe's long neck embodies a rationale for reaching high foliage that no giraffe, no giraffe ancestor, and no evolutionary process ever represented as a goal. The rationale floats free of any intentional attribution; it is real in its effects and real in the sense that it would satisfy a functional description if articulated, but it requires no articulator.

Translated to legal systems, the concept resolves what might be called the intentionality paradox of EPT: how can doctrine evolve through memetic selection when every doctrinal change requires a judge, a scholar, or a legislator to consciously decide? The answer is that conscious decisions operate at the vehicle level; free-floating rationales operate at the replicator level. The Argentine doctrine of ultra-actividad (automatic continuation of collective agreements past expiration) embodies rationales for union organizational stability that no founding judge articulated and that no subsequent court has felt obligated to reconstruct. The doctrine spreads because it reduces the uncertainty costs of collective bargaining for the actors who repeat it, not because anyone grasped this functional logic and willed its propagation. This is free-floating rationale in precise technical sense: the doctrine carries its own reasons without carriers needing to know them.

The implication for EPT is considerable. It means that the CLI does not measure political will or ideological preference; it measures the density of free-floating rationales embedded in the institutional substrate. A CLI of 0.89 for Argentina's labor regime is, in Dennett's terms, an index of how many rationales are floating free in that system: 187 specialized labor courts whose procedural routines embody decades of free-floating doctrine, 3,847 union legal departments whose litigation strategies carry rationales that predate the lawyers who execute them, and curricula in 109 Argentine law schools that reproduce these rationales into each new cohort of practitioners without any member of faculty consciously deciding to transmit them.

2.2 Cranes and Skyhooks: Against Functionalist Lock-in Theory

Dennett's cranes/skyhooks distinction in Darwin's Dangerous Idea targets a specific philosophical error: the invocation of design-from-above to explain complexity that bottom-up processes can generate. A skyhook explains complexity by positing an intelligent designer or transcendent principle that imposes order from outside the causal nexus. A crane is a real lifting device that builds from the substrate up, using materials already generated by prior evolutionary or cultural processes.

In legal theory, the dominant account of institutional persistence is, in Dennett's terms, thoroughly skyhook-committed. Law and economics (Posner, 1973; Priest, 1977) posits that common law rules persist because efficient legislators or judges selected for efficiency from above. New institutional economics (North, 1990) posits that institutions persist because they reduce transaction costs in ways actors recognize and value. Both accounts assume a rational selector that recognizes fitness and translates recognition into selection. Both are skyhooks: they explain persistence by positing a designing intelligence that does not, in fact, exist at the systemic level.

EPT-EGT operates entirely with cranes. The Constitutional Lock-in Index captures crane-built persistence: precedent density (P) is a crane built by citation practices accumulated without anyone deciding to build a lock-in device; doctrinal elaboration (D) is a crane built by academic commentary that reproduces itself through prestige incentives; professional organization (O) is a crane built by bar associations and union legal departments that serve member interests and thereby reinforce the doctrine those interests require; enforcement infrastructure (E) is a crane built by jurisdictional investments that create constituencies for the procedures they implement. None of these cranes required a designing intelligence. Each was built by organisms pursuing local fitness in environments already structured by prior cranes.

The implication for reform analysis is precise and uncomfortable: reform proposals that are themselves skyhooks, rational designs meant to replace the accumulated crane-built structure by legislative fiat, will fail not because they lack support but because they do not provide crane-replacement architecture. The Ley de Bases of 2024 is an instructive case. It attempted to modify substantive labor protections (a skyhook operation: replace doctrine by decree) without dismantling or replacing the crane architecture that produces and reproduces that doctrine. Courts applied pre-Ley de Bases interpretive frameworks because those frameworks are carried by procedural cranes that no legislative text can simply override. This is not judicial activism in any meaningful political sense; it is the crane operating as cranes do.

2.3 Sphexism: Competence Without Comprehension in Legal Practice

In Gödel, Escher, Bach (1979), Douglas Hofstadter described the wasp Sphex ichneumoneus performing a pre-flight inspection routine that becomes pathologically rigid when interrupted: if a cricket is moved while the wasp is inside the burrow, the wasp reemerges, re-inspects, re-enters, and will repeat this cycle indefinitely without adjusting to the obvious futility. Dennett adopted and extended this observation in Elbow Room (1984) and Darwin's Dangerous Idea to describe a broader class of behavioral rigidity: competence without comprehension, routines that execute with high fidelity precisely because they are not subject to reflective interruption.

The concept maps with uncomfortable precision onto Argentine labor jurisprudence. The Corte Suprema's application of the in dubio pro operario doctrine, the doctrine of derechos adquiridos, and the confiscation cap derived from Vizzoti (2004) exhibits exactly the pattern Dennett identifies: high-fidelity execution of routines that no current jurist designed, whose original rationales are no longer reconstructed in opinions, and whose application to novel technological or economic contexts proceeds as if those contexts raised no interpretive question requiring fresh deliberation. This is not negligence or bad faith; it is sphexism as Dennett describes it: the institutional product of socialization processes that select for routine-execution fidelity over adaptive flexibility.

The connection to Heteronomous Bayesian Updating (HBU) is direct. HBU (Lerer, 2026b) describes how legal operators form strong priors through observation of high-authority reactions rather than through direct experience with substantive outcomes. Dennett's sphexism provides the cognitive mechanism: what HBU models as prior-driven blocking, Dennett explains as the behavioral consequence of competence-without-comprehension formation. The law student who observes a senior judge applying ultra-actividad doctrine does not learn why that doctrine is correct; she learns the execution routine. Her future application of the doctrine will exhibit sphexistic fidelity: high competence, absent comprehension, maximum resistance to interruption.

This has a specific and testable implication. Systems with high CLI should show higher rates of sphexistic jurisprudential behavior: opinions that apply doctrine without reconstructing rationale, that treat novel contexts as if they were standard cases, and that resist modification even when explicit legislative instruction contradicts the routine. The expected differential between high-CLI systems (CLI > 0.75) and low-CLI systems (CLI < 0.40) is 40–60 percentage points in the rate of unexplained doctrinal application. This is Prediction 1 of the framework, developed further in Section VI.

III. Integration with HBU and the EGT Typology of Cooperative Intentionality

3.1 The Three-Type Typology and Its Dennettian Enrichment

Prior work (Lerer, 2025c) developed a typology of cooperative intentionality in legal systems drawing on Dennett's intentional stance hierarchy. Type 1 cooperation involves physical-stance actors: parties who interact through direct stimulus-response without modeling others' beliefs. Type 2 involves design-stance actors: parties who model others as rule-followers and cooperate through rule-recognition. Type 3 involves intentional-stance actors: parties who model others' beliefs, goals, and reasons, and cooperate through shared understanding of purpose. Each type generates a distinct class of evolutionary stable strategies (ESS) in the EGT framework.

What Dennett adds to this typology is not a fourth type but a mechanism explaining the transition failures between types. Actors who operate at Type 3 sophistication may nonetheless execute Type 1 routines when institutional design does not support reflective interruption. Sphexism is precisely the mechanism of this downward collapse: the senior partner at a labor law firm has full Type 3 cognitive capacity but applies ultra-actividad doctrine through a Type 1 stimulus-response chain (reform proposal appears: invoke doctrine; doctrine invoked: block reform). The crane architecture of the profession does not reward reflective reconstruction of doctrine rationales; it rewards execution fidelity. The result is Type 3 actors behaving like Type 1 actors through institutionally induced sphexism.

This enriches the EGT predictions in a precise way. In standard game-theoretic analysis, increasing the sophistication of actors should improve cooperation outcomes. In EPT-EGT with Dennett's apparatus, this prediction fails in high-CLI environments: increasing actor sophistication does not improve reform outcomes because the sophisticated actors are executing sphexistic routines that their Type 3 capacity does not interrupt. The empirical prediction is that actor sophistication (measured by legal education levels, judicial prestige, or reform-coalition expertise) should be uncorrelated with reform success in high-CLI systems, while positively correlated in low-CLI systems. This is Prediction 2.

3.2 Free-Floating Rationales and the Dennett-Nash Gap

Prior work (Lerer, 2025d) identified what it termed the Dennett-Nash Gap: the systematic divergence between the ESS predicted by game theory for a given payoff structure and the behavioral equilibrium actually observed in institutional settings. The gap was described empirically but not mechanistically explained: why do actors consistently fail to reach Nash-efficient outcomes even when they have the information and cognitive capacity to do so?

Free-floating rationales provide the mechanism. In a Nash equilibrium analysis, all relevant payoffs are on the table and actors calculate accordingly. But the payoff structure that actors actually perceive is shaped by free-floating rationales embedded in their doctrinal training. An Argentine labor lawyer calculating the payoff of a reform coalition is not calculating over the actual economic consequences of reform; she is calculating over a payoff space structured by decades of free-floating rationales about workers' rights, constitutional hierarchy, and judicial doctrine. The Nash equilibrium in her perceptual payoff space may be very different from the Nash equilibrium in the objective payoff space, and she will converge on the former with high confidence and high fidelity.

The Dennett-Nash Gap is therefore not a cognitive failure in any negative sense. It is the predictable consequence of free-floating rationales shaping the perceived payoff space. This means the gap is not reducible by improving actor information or intelligence; it is reducible only by intervening in the crane architecture that produces and maintains the free-floating rationales. Chile's successful labor reforms of 1990–1993 did not produce more intelligent or better-informed reform actors; they succeeded because the Pinochet-era dismantling of the PBS (professional belief substrate) infrastructure had removed the cranes that would otherwise have anchored free-floating rationales incompatible with reform.

IV. Normative Hysteresis: A Mechanism Beyond Spandrels and Lock-in

4.1 Three Distinct Mechanisms: A Conceptual Map

The EPT-EGT literature, including prior contributions in this program, has deployed three related but conceptually distinct concepts that deserve precise differentiation. Failing to maintain these distinctions produces theoretical confusion with practical consequences for reform diagnosis.

Path dependence, as formalized by Arthur (1989) and applied to institutions by North (1990), describes a system where historical states constrain future trajectories: where you came from determines where you can go. Path dependence is a property of the transition matrix: certain transitions are unavailable given current state and history. The concept is essentially about the topology of the state space.

Institutional lock-in, as captured by the CLI, describes the current rigidity of the system: how resistant the present configuration is to modification by available reform instruments. Lock-in is a cross-sectional measure. A CLI of 0.89 for Argentina describes the system as it stands in 2026 without necessarily specifying how it arrived there or how it would respond to interventions of different intensities.

Normative hysteresis describes something neither concept captures: the cumulative, irreversible deformation that each normative intervention leaves in the institutional substrate, such that the substrate's response to subsequent interventions depends not only on its current state but on the history of interventions it has absorbed. In physical materials science, hysteresis is the property whereby the relationship between applied force and resulting deformation depends on whether force is being applied or removed, and on the prior deformation history of the material. A steel rod that has been plastically deformed is not the same material it was before deformation: its yield strength in subsequent loading cycles differs from that of an undeformed rod, even if the observable geometry is similar.

This distinction is not terminological. Path dependence says the road map has been altered. Lock-in says the current road is blocked. Hysteresis says the material of the road has changed with each vehicle that has traveled it, and the next vehicle will encounter a different substrate than the first. Argentina's 23 failed reform attempts are not 23 tests of the same institutional system. Each attempt has deformed the substrate: courts have produced new doctrine specifically to block that category of reform, unions have developed new organizational responses to that type of legislative challenge, law schools have trained cohorts whose practice knowledge includes the failure of those attempts. The system that faces a 24th reform proposal in 2026 is measurably more deformed than the one that faced the first in 1990.

4.2 Hysteresis and the Architecture of Spandrels

The relationship between normative hysteresis and Gould's spandrels (enjutas) is more precise than an analogy. In Gould and Lewontin's original formulation (1979), spandrels are the spaces between a rounded arch and a rectangular surround: they are not selected for, they arise as architectural necessities from the design choice that does produce the arch, and they can subsequently be co-opted for decorative or functional purposes (a process Gould and Vrba would later term exaptation). Spandrels are, in the language of the framework, structural invariants: they persist because of the architecture, not because they contribute to fitness.

Hysteresis is the mechanism by which spandrels accumulate. A single legislative intervention generates spandrels: the specialized courts required to apply the new rule, the academic commentary that the rule generates, the professional networks that the commentary connects. These are not selected for the new rule's purposes; they are architectural byproducts. But they do not remain static. Subsequent interventions add to, interact with, and modify these spandrels. The spandrels generated by the 1957 constitutional insertion of Article 14bis were compounded by the 1974 Ley de Contrato de Trabajo, then by the 1994 treaty elevation of ILO Conventions 87 and 98, then by the Vizzoti precedent of 2004. Each intervention generated new spandrels; those spandrels interacted with existing ones; and the resulting architectural complexity is not the simple sum of the parts. This is hysteretic accumulation: the material properties of the substrate have changed with each deformation cycle.

The normatively important consequence is asymmetry. Repealing Article 14bis would not return the institutional substrate to its 1956 state any more than removing a force from a plastically deformed steel rod returns the rod to its original geometry. The spandrels generated by 68 years of Article 14bis architecture, the 187 specialized labor courts, the 3,847 union legal departments, the citation networks that have incorporated its doctrine into thousands of subsequent rulings, would persist past repeal. The Regulatory Hysteresis Index (IHR) for Argentina's labor regime, calculated at 0.91, is precisely this measurement: it captures the depth of the substrate deformation independent of what the current rules say. Chile's IHR of approximately 0.31 reflects not just lower lock-in but genuinely lower substrate deformation: the Pinochet-era dismantling was a catastrophic deformation in the reverse direction, analogous to a material being re-annealed rather than simply returned to lower stress.

4.3 Dennett's Free-Floating Rationales as Hysteretic Residue

The connection between normative hysteresis and Dennett's apparatus is, on reflection, the most conceptually potent element of the integration. Free-floating rationales are not only the mechanism by which doctrine propagates without conscious transmission; they are also the primary content of hysteretic residue. When a norm is repealed, its free-floating rationales do not repeal with it.

Consider the ultra-actividad doctrine in Argentine labor law. This doctrine, according to which collective agreements continue in force indefinitely after expiration until replaced by a new agreement, was developed judicially between the 1970s and the 2000s without explicit constitutional mandate. It embodies rationales for union organizational stability and worker protection that no court has explicitly articulated in doctrinal terms: it floats free of any authorial intention. If the legislative basis for collective bargaining were modified tomorrow, the free-floating rationales embedded in the ultra-actividad doctrine would not evaporate. They are carried in the procedural habits of 187 specialized courts, in the strategic calculations of 3,847 union legal departments, and in the doctrinal training of every labor lawyer graduated in Argentina since approximately 1985. The hysteretic residue of the doctrine would persist for at least a generation past its formal repeal.

This is the precise sense in which sphexism is the behavioral face of normative hysteresis. The competence-without-comprehension that Dennett identifies in sphexistic actors is the behavioral expression of hysteretic deformation: the operator executes the routine because the substrate was deformed in a specific direction by prior normative cycles, and the deformation persists independently of whether the originating norm is still in force. Sphexism is not merely cognitive conservatism; it is the behavioral signature of a substrate that has not returned to its pre-deformation state.

4.4 The Regulatory Hysteresis Index and Its Relationship to CLI

The CLI and the Regulatory Hysteresis Index (IHR) are correlated but measure distinct phenomena. $CLI = 0.30P + 0.25D + 0.20O + 0.25E$ captures the current density of institutional lock-in mechanisms: precedent density (P), doctrinal elaboration (D), professional organization (O), and enforcement infrastructure (E). It is a cross-sectional measure of resistance at a given moment.

The IHR captures the depth of hysteretic deformation: the degree to which the institutional substrate has been permanently altered by the history of normative interventions, such that its response to new interventions differs from the response of an undeformed substrate. An IHR is calculated by integrating, over time, the weighted sum of normative interventions that have left measurable structural residues. For Argentina's labor regime, this includes the original 1957 constitutional inscription, the 1974 Ley de Contrato de Trabajo, the 1994 treaty elevation, and the accumulated judicial doctrine, yielding IHR = 0.91. For Chile's post-1990 labor regime, the Pinochet-era dismantling constitutes a large negative deformation that reduces IHR to approximately 0.31 despite a subsequent accumulation of new doctrine.

The two indices make distinct predictions. CLI predicts the probability of reform success in the next attempt, given current institutional configuration. IHR predicts the trajectory of reform difficulty across multiple attempts: in a high-IHR system, each failed attempt increases CLI (through new doctrine, new professional entrenchment, new precedential reinforcement), while each successful attempt reduces IHR only partially, because the structural residues of prior cycles do not fully reverse. This is Prediction 3: in systems with equivalent current CLI, those with higher IHR should show steeper CLI increases per failed reform attempt, and shallower CLI decreases per successful one.

Table 1: Conceptual Comparison of Three Distinct Mechanisms

Mechanism	What it measures	Temporal nature	Reversibility	EPT equivalent
Path dependence	State space topology: which transitions are available	Historical	Partial: some paths can be re-opened	Founding document constraints
Lock-in (CLI)	Current resistance to available reform instruments	Cross-sectional	Possible with sufficient exogenous shock	CLI composite score
Normative hysteresis (IHR)	Accumulated substrate deformation from intervention history	Longitudinal/cumulative	Asymmetric: return path differs from departure path	IHR composite score
Spandrels	Architectural byproducts without direct selective function	Structural/static	Resistant but not impossible to eliminate	Incidental institutional infrastructure

V. Parallelism, Convergence, and the Failure of Legal Transplants

5.1 Two Mechanisms of Institutional Similarity

The distinction between evolutionary parallelism and convergence, articulated in contemporary evo-devo biology (Wake, 1991; Arendt and Reznick, 2008), provides a diagnostic tool for a problem that EPT has described empirically but not mechanistically explained: why do legal transplants fail systematically, even when functional pressures should favor their adoption?

Convergence occurs when phylogenetically distant lineages develop similar features through independent responses to similar selective pressures. The classic case is the eye: cephalopods, vertebrates, and arthropods developed independently functional optical organs without shared developmental ancestry. The similarity is functional: common external pressure (light-based predator-prey dynamics) drove similar solutions from different starting points. Convergence is the prediction of pure functional selectionism.

Parallelism occurs when phylogenetically related lineages develop similar features because they share deep homologous developmental generators. The Pax-6 regulatory gene, which is so deeply conserved that a vertebrate version can induce ectopic eyes in Drosophila, produces similar developmental outcomes not because of common selective pressure but because of shared genetic regulatory architecture. Parallelism is the prediction of structural-developmental theory.

Applied to legal systems, the distinction separates two classes of institutional similarity that functionalist theory conflates. Countries with similar economic development levels or similar political pressures may converge on similar institutional solutions through functional selection: common law and civil law systems have both developed commercial arbitration frameworks, constitutional courts, and consumer protection regimes in response to similar economic pressures. This is convergence. But Hispanic-American countries show a distinct pattern: they share labor law architectures not because they all independently solved the same functional problem but because they share deep homologous regulatory generators inherited from the Napoleonic civil law tradition, Catholic social corporatism, and ILO conventions integrated at constitutional level. This is parallelism.

5.2 The Constitutional Pax-6 and the Generator Incompatibility Hypothesis

The generator incompatibility hypothesis holds that legal transplants fail not primarily because of political resistance but because the recipient system lacks the deep homologous generators that the transplant presupposes. A transplant that succeeded in a system with Pax-6-equivalent constitutional architecture will fail in a system without it, regardless of functional pressure.

In constitutional legal systems, the deep homologous generators are three: the constitutional amendment procedure (which determines what level of intervention can modify the basic architecture), the jurisdictional architecture of constitutional review (which determines which actors can convert functional pressure into doctrinal change), and the professional socialization curriculum (which determines what rationales are transmitted through the crane architecture of legal education). These three together constitute what might be called the constitutional Pax-6: the deep regulatory network that channels legal evolution regardless of surface functional pressures.

Argentina and Chile share superficially similar constitutional architectures, both civil law systems with democratic governments and comparable economic structures. But their constitutional Pax-6 profiles are radically different. Argentina's amendment procedure requires two-thirds majorities in both chambers plus a constituent convention; its constitutional review is concentrated in a Supreme Court with strong stare decisis norms; its professional socialization curriculum emphasizes constitutional social rights as non-derogable fundamental values. Chile's amendment procedure post-1990 requires lower supermajorities; its constitutional review is distributed; its professional socialization post-1990 incorporates the labor code reform as a legitimate precedent rather than an exception. A labor reform transplant that worked in Chile in 1990–1993 cannot be expected to work in Argentina, not because Argentine workers prefer the current system, but because the constitutional Pax-6 of Argentina lacks the deep generators that made the Chilean transplant viable.

This explains what the IusMorfos tool (Lerer, 2025e) captures empirically in its transplant compatibility scores. IusMorfos weights compatibility along five dimensions; the generator incompatibility hypothesis predicts that the three deepest dimensions (amendment architecture, review jurisdiction, and socialization curriculum) should carry substantially more predictive weight than the two surface dimensions (legislative majority and judicial activism levels). Prediction 4: in transplant attempts where deep generator compatibility is high (score > 0.70 on the three deep dimensions), success rates should be comparable to the source system's baseline; where deep generator compatibility is low (score < 0.40), success rates should be near zero regardless of surface political conditions.

Table 2: Parallelism vs. Convergence in Latin American Labor Law Systems

Country	CLI	IHR (est.)	Shared generators	Similarity type	Reform outcome
Argentina	0.89	0.91	Napoleonic code + Catholic corporatism + ILO hierarchy	Parallelism with Brazil, Mexico	0/23 sustained
Brazil	0.40	0.55	CLT 1943 (Vargas era), partial Napoleonic	Partial parallelism with Argentina	2017 reform (partial)
Mexico	0.70	0.68	Art. 123 (1917) + revolutionary corporatism	Parallelism with Argentina (different branch)	Limited reform
Chile	0.24	0.31	Post-1990 Labor Code; prior generators dismantled	Divergent (catastrophic deformation)	6/6 reforms 1990-2000
Spain	0.51	0.48	Civil law + EU pressure (external generator)	Convergence (EU functional pressure)	5/6 reforms 1994-2022

VI. Five Falsifiable Predictions

The integrated framework generates five predictions that are independently falsifiable against available data, each testing a distinct mechanism.

Prediction 1: The Sphexism Rate Differential

In high-CLI systems (CLI > 0.75), the rate of unexplained doctrinal application (opinions applying doctrine without rationale reconstruction, henceforth the sphexism rate) should exceed the rate in low-CLI systems (CLI < 0.40) by 40–60 percentage points. Measurement protocol: code 500 labor opinions from each system in the dataset (Argentina, Chile, Brazil, Spain) for the presence or absence of explicit rationale reconstruction in each doctrinal application. Expected result: Argentina ~75–85% sphexism rate; Chile ~20–30%.

Prediction 2: The Sophistication-Outcome Decoupling

In high-CLI systems, actor sophistication (measured by judicial prestige rankings, reform-coalition legal expertise, or legislative committee expertise levels) should be statistically uncorrelated with reform outcome. In low-CLI systems, sophistication should be positively correlated. This prediction distinguishes EPT-EGT from standard public choice theory, which predicts sophistication-outcome correlation regardless of institutional context.

Prediction 3: The Hysteretic Accumulation Curve

In high-IHR systems, each failed reform attempt should generate a measurable increase in CLI components (particularly D and O), producing a positive feedback dynamic: failure increases lock-in, increased lock-in makes future failures more likely. The IHR-CLI trajectory across Argentina's 23 reform attempts should show a statistically significant upward trend in CLI over the 1990–2025 period, controlling for political cycle. In Chile post-1990, the inverse relationship should hold: successful reforms should show decreasing IHR over time, but the decrease should be slower than the increase generated by a comparable number of failed attempts in Argentina.

Prediction 4: The Generator Compatibility Threshold

Legal transplant success rates should show a threshold effect at deep-generator compatibility score of approximately 0.60 (measured on the three-dimension IusMorfos deep generator scale): transplants above this threshold should succeed at rates comparable to the source system's baseline; transplants below this threshold should fail at rates exceeding 80%, regardless of surface political conditions, crisis context, or coalition size.

Prediction 5: The Reform Crane Gap

Reforms that modify the target doctrine without building replacement crane architecture (reforms that attempt to change doctrine by legislative text without modifying the enforcement infrastructure, professional organization, or judicial appointment architecture that reproduces the doctrine) should show survival rates below 25% at five years. This prediction tests the cranes/skyhooks distinction: skyhook-type reforms, those that attempt to impose change from above without crane-building, should fail at a rate predictable from the IHR of the target domain, not from political conditions.

Table 3: Five Falsifiable Predictions and Test Protocols

#	Prediction	Data source	Falsification criterion
P1	Sphexism rate 40-60 pts higher in CLI>0.75 systems	500 opinions per country, coded for rationale reconstruction	No significant differential between high/low CLI groups
P2	Actor sophistication decoupled from reform success in high-CLI	Reform dataset 60 cases + actor expertise measures	Significant positive sophistication-outcome correlation in high-CLI systems
P3	Each failed attempt increases CLI components D and O	CLI component time series, Argentina 1990-2025	No statistically significant upward trend in D, O over reform attempts
P4	Generator compatibility threshold at 0.60 for transplant success	IusMorfos scores + transplant outcome database	No threshold effect; success rates linear with compatibility score

#	Prediction	Data source	Falsification criterion
P5	Reforms without crane replacement < 25% survival at 5 years	23 Argentine reforms coded for crane replacement presence	Crane-less reforms survive at comparable rates to crane-building reforms

VII. Application: Argentina 2024–2025 as Natural Experiment

The Ley de Bases y Puntos de Partida para la Libertad de los Argentinos, approved by Congress in June 2024, provides a near-ideal natural experiment for the integrated framework. The law attempted significant labor deregulation within a high-CLI (0.89), high-IHR (0.91) institutional substrate, under conditions of severe economic emergency, with a reforming coalition of unusual political strength for Argentine standards.

The framework predicts, and the record confirms, the following pattern. First, provisions that modified surface doctrine without altering crane architecture (the RIGI investment incentives, the privatization provisions) showed initial legislative success followed by rapid judicial containment through free-floating rationales embedded in the administrative law doctrinal network. Second, provisions that directly targeted the crane components of labor lock-in (union registration reform, collective bargaining scope modification) were invalidated by the CSJN through Vizzoti-era doctrine applied without explicit rationale reconstruction: Prediction 1 behavior. Third, the crisis context, which standard political economy theory predicts should increase reform probability, generated the opposite effect for the labor provisions: economic emergency activated the free-floating rationale network defending acquired rights, increasing institutional resistance rather than reducing it. This is consistent with the HBU model (crisis generates prediction error for actors with strong priors, triggering active inference rather than perceptual inference) and with prior empirical findings (Lerer, 2026a) that crisis context has a coefficient of 2.1 in the reform success logistic regression, paradoxically increasing failure probability in high-CLI systems.

The Ley de Bases case also illustrates the generator incompatibility dynamic. The reform was designed by economists and political scientists drawing on the Chilean 1990–1993 and Spanish 1994–1997 reform experiences. Both source systems have deep generator profiles incompatible with Argentina's constitutional Pax-6: Chile had dismantled its PBS during the military period; Spain was operating under EU functional pressure that constituted an external crane supplementing its domestic institutional capacity. Transplanting their reform texts into Argentina's generator architecture was, in IusMorfos terms, an attempt to express a gene in a regulatory environment that lacks the necessary transcription factors. The transplant was not merely politically difficult; it was architecturally incompatible.

The hysteresis consequence of the 2024 attempt is still accumulating. New CSJN doctrine produced to contain Ley de Bases provisions will become free-floating rationale in future cases. New union organizational capacity developed to block those provisions will remain as crane architecture for future blocking. The IHR of Argentina's labor regime on March 2026 is measurably higher than it was in June 2024, not because the reform failed in the political sense but because the failure generated hysteretic deformation in the direction of increased lock-in.

VIII. Literature Review and Theoretical Positioning

8.1 The Memetics Debate: Fracchia, Lewontin, and Hull

The application of memetic theory to social institutions has generated sustained methodological criticism. Fracchia and Lewontin (1999) argued that cultural evolution by natural selection is either a tautology (if fitness is defined post-hoc by survival) or an empirical claim that requires independently measurable fitness, which cultural theory has not provided. The EPT-EGT framework takes this criticism seriously. The CLI provides independently measurable fitness proxies (citation density, doctrinal elaboration, professional organization density, enforcement infrastructure depth) that allow prediction prior to outcome. The 78% predictive accuracy of CLI in the 60-case dataset is not a tautological result: CLI was constructed from doctrinal and institutional data without using reform outcome as an input.

Lewens (2007) raised a further objection: even if cultural entities replicate, they do not replicate with the fidelity required for Darwinian selection to operate. Legal doctrines mutate rapidly, are deliberately modified by their carriers, and do not have the parent-offspring transmission fidelity that evolutionary models require. The response lies precisely in the crane architecture identified in this paper. The sphexism of legal operators, the free-floating rationales embedded in professional training, and the hysteretic residue of prior normative cycles together produce fidelity that individual conscious decision-making could not. The Argentine legal system does not produce high doctrinal fidelity because each individual jurist wills fidelity; it produces fidelity because the crane architecture makes deviation cognitively and professionally costly.

Hull's (1988) replicator-interactor distinction provides the cleanest alignment with EPT. Legal memes are Hull-type replicators: entities that pass on their structure through replication. Legal institutions are Hull-type interactors: entities that interact with their environment, producing differential replication of the replicators they carry. The CLI measures interactor architecture; the IHR measures the accumulated effect of interactor history on replicator fidelity. Dennett's free-floating rationales describe how replicators preserve structure across generations of interactors without requiring that interactors consciously represent the structure they carry.

8.2 Positioning Relative to New Institutionalism and Historical Institutionalism

The EPT-EGT framework occupies a specific position in relation to the major schools of institutional theory. North's (1990) transaction cost institutionalism explains persistence through efficiency: institutions persist because they reduce coordination costs that actors recognize and value. This is a skyhook account in Dennett's sense, and the 23-attempt failure record of Argentine labor reform, in a system that virtually all economic analysts agree is inefficient, constitutes its empirical refutation in this domain.

Pierson's (2004) path dependence approach comes closer but stops short of the mechanism. Pierson documents that historical sequences generate increasing returns that make change costly; he does not explain why actors who recognize the inefficiency and have the political resources to change it nonetheless fail. HBU, Dennett's sphexism, and normative hysteresis together explain the residual: actors who recognize inefficiency are nonetheless executing sphexistic routines through a substrate deformed by hysteretic cycles, and the free-floating rationales embedded in that substrate shape their perceived payoff space in ways that make the efficient change invisible as an option.

Mahoney's (2000) work on path dependence in Central America is the most direct precursor, identifying what he calls utilitarian, functional, power, and legitimation lock-in mechanisms. The EPT-EGT framework subsumes all four under a single evolutionary mechanism: they are all forms of crane architecture that maintains free-floating rationales incompatible with reform. The IHR integrates across these mechanisms rather than treating them as four separate causal stories.

IX. Positive Cases: Colombia 1991 and Chile 1990–1993

The framework's predictive power requires not only explaining failure in high-IHR systems but also explaining success in systems where crane-building preceded reform.

Colombia's 1991 constitutional reform constitutes a natural experiment in crane-building prior to doctrinal change. The Asamblea Nacional Constituyente effectively built new crane architecture before attempting normative change: it created a new constitutional court with distinct jurisdictional foundations, reformed the public defender system, and modified the professional socialization environment by requiring new constitutional law courses in all law schools. The generator incompatibility problem was addressed not by transplanting foreign doctrine but by building domestic generators capable of producing compatible doctrine. The resulting CLI for Colombian constitutional law dropped from approximately 0.72 (under the 1886 constitution) to 0.41 within ten years of the 1991 reform, with corresponding improvement in reform success rates for constitutional challenges.

Chile's 1990–1993 labor reforms succeeded because the Pinochet-era dismantling had operated as a hysteretic deformation in the reverse direction. The PBS infrastructure of the pre-1973 labor regime (socialist union networks, progressive labor doctrine, strong court enforcement of collective rights) had been destroyed rather than merely blocked. This constitutes a catastrophic deformation analogous to material re-annealing: the substrate was reset to a lower hysteretic state, reducing IHR from an estimated 0.85 in 1973 to approximately 0.31 by 1990. The Aylwin government's labor reforms did not need to overcome accumulated hysteretic deformation; they were introducing new cranes into a substrate that, through traumatic transformation, had become receptive. This is a positive prediction of the framework: not all catastrophic shocks generate reform windows (the shock must target O and E components specifically, not merely P or D), but shocks that genuinely reduce IHR, whatever their normative character, create reform windows that rational-actor models cannot predict.

X. Conclusion: Synthesis and Agenda

The integrated framework advances the EPT-EGT research program in three directions. Conceptually, Dennett closes the intentionality gap that the prior program's empirical success could not by itself resolve: deliberating actors and memetic selection are compatible because deliberation operates at the vehicle level while free-floating rationales operate at the replicator level, and sphexism describes the mechanism by which crane architecture produces fidelity without conscious transmission. Theoretically, normative hysteresis is introduced as a distinct mechanism from path dependence, lock-in, and spandrels: it captures the accumulated, asymmetric deformation of the institutional substrate across intervention cycles, generating the prediction that each failed reform attempt increases future resistance through IHR accumulation. Methodologically, the parallelism/convergence distinction from evo-devo separates two classes of institutional similarity that functionalism conflates, generating the generator incompatibility hypothesis as a diagnostic for legal transplant failure.

Five falsifiable predictions distinguish these contributions from prior work. The sphexism rate differential (P1) is testable against existing court records. The sophistication-outcome decoupling (P2) tests the EPT explanation against public choice alternatives. The hysteretic accumulation curve (P3) tests whether failed attempts increase lock-in in the predicted IHR-CLI trajectory. The generator compatibility threshold (P4) tests the transplant failure mechanism. The reform crane gap (P5) tests the cranes/skyhooks distinction in the most direct possible way.

The program's immediate agenda is twofold. First, operationalization of the IHR for the six-country dataset, using the protocol described in the Appendix, to enable cross-national testing of P3. Second, application of the generator compatibility scoring to the IusMorfos database to test P4 against the existing 60-case record. If both predictions survive, the framework will have moved from an empirically well-supported account of why reform fails to a mechanistically grounded theory of the conditions under which it can succeed.

The practical implication is diagnostic: before attempting any reform in a high-CLI, high-IHR domain, a reform strategy should begin with IHR measurement, generator compatibility scoring, and crane architecture analysis. Not because these tools guarantee success, but because their absence guarantees the kind of failure that looks inexplicable to standard political analysis and is entirely predictable to evolutionary institutional theory.

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***AI Use Disclosure**: The quantitative values reported in this paper — Constitutional Lock-in Index (CLI) scores, Regulatory Hysteresis Index (IHR) estimates, and predictive accuracy figures — represent theoretical constructs calibrated against available institutional data rather than outputs of fully independent empirical validation. CLI scores for Argentina (0.89), Chile (0.24), Brazil (0.40), and Spain (0.51) derive from the methodology described in Lerer (2026a) and are consistent across prior publications in this research program. IHR estimates are preliminary and should be treated as ordinal approximations pending the operationalization protocol described in the Appendix. Readers should interpret all numerical claims as theoretically motivated estimations with the confidence levels appropriate to an early-stage research program, not as definitive empirical measurements. The falsifiable predictions in Section VI are intended precisely to subject these estimates to independent testing.*

Appendix: Operationalization Protocol for the IHR

The Regulatory Hysteresis Index (IHR) is calculated as the weighted time-integral of normative interventions that have left measurable structural residues, using the following formula:

$IHR = \Sigma\, [w_i \times R_i \times (1 - D_i)] / T$

Where: w_i = weight of intervention i (based on constitutional rank: 1.0 for constitutional text, 0.8 for international treaty with constitutional rank, 0.6 for organic law, 0.4 for ordinary legislation, 0.2 for judicial doctrine); R_i = measured residue intensity of intervention i (estimated from: number of specialized institutions created, volume of derivative doctrine generated, number of practitioners whose practice depends on the intervention); D_i = observed decay rate of residue i since intervention (estimated from institutional dismantling events); T = normalization constant.

Appendix Table: IHR Estimation Protocol by Country

Country	CLI (2026)	IHR (est.)	Dominant deformation source	Predicted CLI trajectory
Argentina	0.89	0.91	1957 constitutional + 1974 LCT + 1994 treaty elevation + Vizzoti 2004	Rising: each failed attempt +0.02–0.04 CLI increment
Brazil	0.40	0.55	CLT 1943 + slow doctrinal accumulation	Stable: 2017 reform reduced D; reversal risk
Chile	0.24	0.31	Post-1990 accumulation on reset substrate	Slow rise: new crane-building post-Boric reforms
Spain	0.51	0.48	EU-driven crane replacement partially offset domestic IHR	Moderate: EU constraints limit domestic IHR growth
Mexico	0.70	0.68	Art. 123 + corporate union architecture	Rising: 2019 labor reform added new crane layer
Colombia	0.41	0.39	1991 constitutional reset reduced prior IHR substantially	Stable to declining: 1991 crane-building effects persist

The sphexism rate operationalization protocol proceeds as follows. For a given jurisdiction and five-year period, draw a stratified random sample of 500 opinions from the target doctrinal domain (labor law, constitutional law, as appropriate). For each opinion, identify all applications of prior doctrine. Code each application as (1) with explicit rationale reconstruction: the opinion articulates why the doctrine applies to the current facts, engaging with any novel elements; (2) with implicit but traceable rationale: the application follows established case categories without novel engagement but the rationale is reconstructable from the opinion text; or (3) without reconstructable rationale: the doctrine is applied as automatic rule with no engagement with the facts' alignment with the rationale. Category (3) applications constitute the sphexism rate. Expected values: Argentina > 70%; Chile < 30%; Brazil 45–55%; Spain 35–45%. These predictions are falsifiable by any sufficiently large sample of opinions from these jurisdictions.